

Log log plot: Estimating Critical Exponents

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Summary

Background (physicists' picture)

- ➊ Phase transitions in percolation
- ➋ Critical exponents, scaling, universality
- ➌ From physics heuristic to statistical question

This work (joint with A. Teixeira and R. Imbuzeiro de Oliveira)

- ➍ The log-log estimator and its three-part split
- ➎ Main theorem: CLT for $\hat{\gamma}$
- ➏ Lower bound and optimality
- ➐ m_0 choices empirically
- ➑ Future works

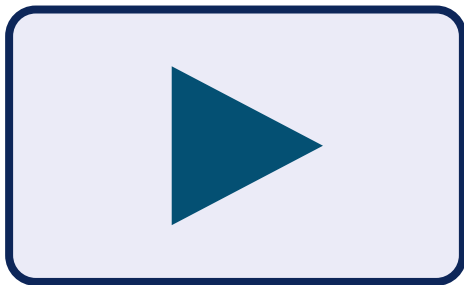
PERCOLATION PROCESSES

I. CRYSTALS AND MAZES

BY S. R. BROADBENT AND J. M. HAMMERSLEY

Received 15 August 1956

ABSTRACT. The paper studies, in a general way, how the random properties of a 'medium' influence the percolation of a 'fluid' through it. The treatment differs from conventional diffusion theory, in which it is the random properties of the fluid that matter. Fluid and medium bear general interpretations: for example, solute diffusing through solvent, electrons migrating over an atomic lattice, molecules penetrating a porous solid, disease infecting a community, etc.



SIMULATION

`run - perc_interactive.py`

based on the book [Malthe-Sørenssen'24]

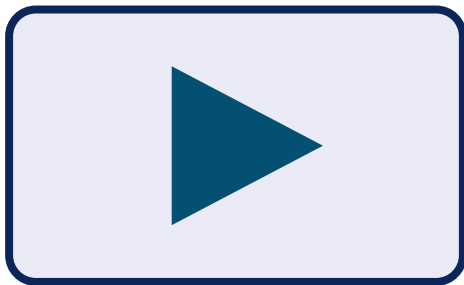
Phase transitions everywhere

- Ising model (magnetisation)
- XY model
- Diffusion / interacting particle systems
- Random walks
- Spread of infection
- Polymers / self-avoiding walks

Common thread: a control parameter, an order parameter, a critical value, power-law behaviour at the threshold.

Simulating percolation on a computer

Uniform coupling. Draw one i.i.d. array $U_x \sim \text{Unif}[0, 1]$ once; for any p , declare site x open iff $U_x \leq p$. Same realisation, every p simultaneously.



GIF

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run - lattices_merged.gif
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based on the book [Malthe-Sørenssen'24]

Defining the critical parameter p_c

Fix a lattice. Let

$$\theta(p) = \mathbb{P}_p(0 \leftrightarrow \infty)$$

be the probability that the cluster of the origin is infinite. θ is monotone non-decreasing in p , and

$$p_c := \sup\{p \in [0, 1] : \theta(p) = 0\} = \inf\{p \in [0, 1] : \theta(p) > 0\}.$$

- $p < p_c$: with probability 1, every cluster is finite (sub-critical phase).
- $p > p_c$: a unique infinite cluster exists almost surely (super-critical phase).
- $p = p_c$: **critical phase**. No infinite cluster, but clusters of every scale — the scale-invariant regime where $\mathbb{E}V(r) \sim r^{d_f}$ lives.

lattice	honeycomb	square	triangular
vertex degree	3	4	6
p_c	≈ 0.6962	≈ 0.59274621	$= 0.5$

The zoo of critical exponents

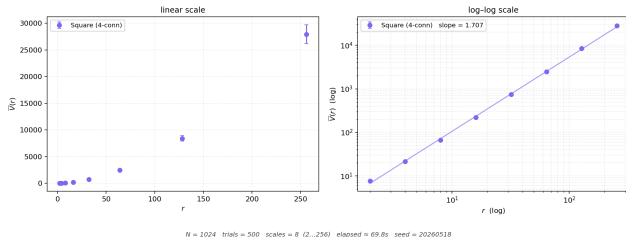
Function		Behavior	Exp.	Scaling rel.
► Cluster volume at p_c	$V(r)$	$\mathbb{E}V(r) \sim r^{d_f}$	d_f	$d_f = d/(\tau - 1)$
Cluster number density	$n(s, p)$	$n(s, p) \sim s^{-\tau} f(s p - p_c ^{1/\sigma})$	(τ, σ)	—
Percolation probability	$\theta(p)$	$\theta(p) \sim (p - p_c)^\beta$	β	$\beta = (\tau - 2)/\sigma$
► Mean cluster size	$\chi^f(p)$	$\chi^f(p) \sim p - p_c ^{-\gamma}$	γ	$\gamma = (3 - \tau)/\sigma$
Number of clusters per vertex	$\kappa(p)$	$\kappa(p) \sim p - p_c ^{2-\alpha}$	α	$\alpha = 2 - (\tau - 1)/\sigma$
Cluster moments	$\chi_k^f(p)$	$\chi_{k+1}^f(p)/\chi_k^f(p) \sim p - p_c ^{-\Delta}$	Δ	$\Delta = 1/\sigma = \gamma + \beta$

Two anchor observables

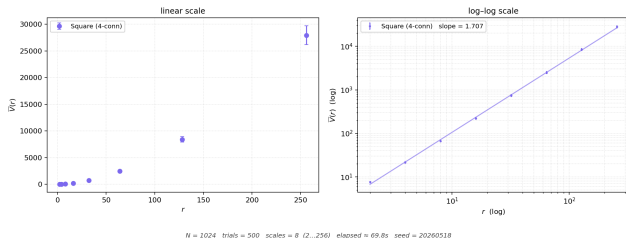
- (a) cluster volume $\mathbb{E}V(r) \sim r^{d_f}$ at $p = p_c$.
- (b) mean cluster size $\chi(p) \sim |p - p_c|^{-\gamma}$ (off-critical).

log-log linearises a power law (& finite-size effects)

Square (4-conn) — cluster volume vs. box size at p_c (asymptotic theory: $d_f = 1.8958$)



Square (4-conn) — cluster volume vs. box size at p_c (asymptotic theory: $d_f = 1.8958$)



Heuristic:

$$\mathbb{E}V(r) \sim r^{d_f}$$

$$\Rightarrow \log \mathbb{E}V(r) \approx d_f \log r + \text{const.}$$

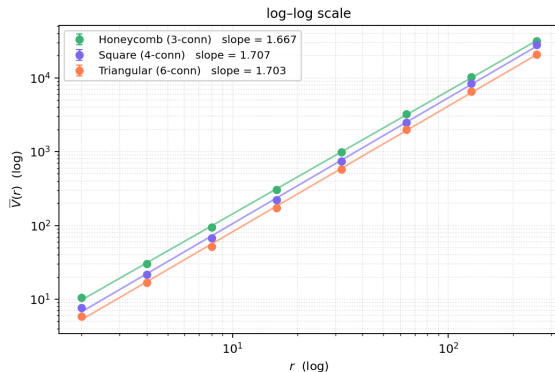
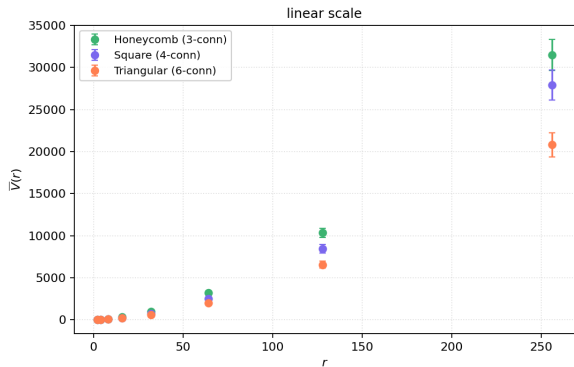
Physicists “read the slope”. But the relation is only **asymptotic** — at small r there are corrections.

We call this error for small r **Finite-size effect**.

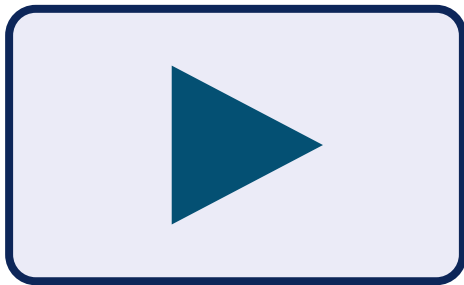
Remark. The standard fix is to **drop the first m_0 scales** and fit the OLS line on the remaining m points. m_0 becomes the main design knob.

Universality across lattices

Fractal dimension at p_c — universality across lattices (theory: $d_f = 91/48 \approx 1.8958$)



$N = 1024$ trials = 500 scales = 8 (2...256) elapsed $\approx 69.8s$ seed = 20260518



GIF

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run - percolation_meanfield.gif
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What physicists measure vs. what they want

Want (unobservable)

$$\mathbb{E}V(r)$$

the population mean

Have (observable)

$$\overline{V}(r) = \frac{1}{n} \sum_{t=1}^n V_t(r)$$

sample mean over n independent trials

Two reasons our log-log reading can be off:

- (a) $\overline{V}(r) \neq \mathbb{E}V(r)$ — random fluctuations, source of **variance**;
- (b) $\mathbb{E}V(r)$ is only **asymptotically** a power law — deterministic offset at small r , source of **bias**.

The next two frames formalise each source.

Setup: a generic observable Y_i

Let Y_i be a generic observable with γ as the related critical exponent.

(A0) $Y_i > 0$ (so $\log Y_i$ is defined).

(A1) Assumption 1.

$$\mathbb{E} Y_i = a_0 i^\gamma \exp\left(a_1 i^{-1} + \phi(i) i^{-2}\right), \quad \phi \text{ bounded.}$$

Identification for our running example: $Y_i \leftrightarrow V(r)$ with $i = r$, $\gamma \leftrightarrow d_f$.

Reading the expansion.

- $a_0 i^\gamma$ — the power law we are after.
- a_1/i — leading finite-size correction (boundary, discreteness). Its log is the source of **deterministic bias** at small i .
- $\phi(i)/i^2$ — bounded remainder, one power faster.
- This is **not** exponential decay: the exponential is just a tidy way to write the multiplicative correction $a_0 i^\gamma (1 + a_1/i + \dots)$ in a form whose log is additive.

Chalkboard: per-scale decomposition

Define the random ratio $\bar{\xi}_k := \bar{Y}_{\rho^k} / \mathbb{E} Y_{\rho^k}$ (mean 1).

We can decompose our log-log plot as follows: write $\log \bar{Y}_{\rho^k} = \log \mathbb{E} Y_{\rho^k} + \log \bar{\xi}_k$ and substitute $i = \rho^k$ in the expansion of the previous slide.

$$\log \bar{Y}_{\rho^k} = \underbrace{\log a_0 + k \gamma \log \rho}_{\text{signal (linear in } k\text{)}} + \underbrace{a_1 \rho^{-k} + \phi(\rho^k) \rho^{-2k}}_{\text{finite-size bias (deterministic)}} + \underbrace{\log \bar{\xi}_k}_{\text{random}}.$$

Notation. Write $\alpha = \log a_0$ and $\beta = \gamma \log \rho$ for the signal coefficients.

- Signal: linear in k with slope $\beta = \gamma \log \rho$.
- Bias: decays **geometrically** in k (since $\rho > 1$). After a few scales it sits below statistical noise.
- Random part: $\mathbb{E} \log \bar{\xi}_k \approx 0$, $\text{Var} \log \bar{\xi}_k \asymp \sigma_k^2/n$.

Bias–variance trade-off

Every term in the per-scale decomposition is now formal:

- the finite-size bias has size $\asymp \rho^{-k}$ (deterministic, shrinks with k);
- the random part has size $\asymp \sigma_k/\sqrt{n}$ (stochastic, controlled by simulation cost).

Small r (cheap)

- ✓ many trials affordable in a fixed compute budget — **low variance** of $\overline{V}(r)$.
- × finite-size correction a_1/r is large — **high bias**.

Large r (expensive)

- ✓ asymptotic regime: $a_1/r \rightarrow 0$ — **low bias**.
- × each trial costs more time, fewer trials fit in the budget — **high variance**.

Choosing which scales to use is a **statistical decision**, not an aesthetic one.

*The bias–variance picture above is common knowledge in the physics literature; the **compute-budget formalisation** that pins (m_0, m, n) against a single resource constraint B is, to our knowledge, new.*

From here on: our contribution

joint with A. Teixeira and R. Imbuzeiro de Oliveira

- 5 Section 5 — a CLT for $\hat{\gamma}$
- 6 Section 6 — minimax lower bound — the log-log plot is near-optimal
- 7 Section 7 — m_0 choices empirically

Linear regression

Data: $(k, \log \bar{Y}_{\rho^k})$ for $k = m_0 + 1, \dots, m_0 + m$.

The **ordinary least squares** (OLS) slope has the closed form

$$\hat{\beta} = \sum_{k=m_0+1}^{m_0+m} w_{k,m} \log \bar{Y}_{\rho^k}, \quad w_{k,m} = \frac{12 (k - m_0 - (m+1)/2)}{m(m^2 - 1)}.$$

Hence the estimator of the target exponent:

$$\hat{\gamma} = \frac{\hat{\beta}}{\log \rho}.$$

Linear regression as a linear functional

Recall $\hat{\beta} = \sum_k w_{k,m} \log \bar{Y}_{\rho^k}$ — a **deterministic linear combination** of the $\log \bar{Y}_{\rho^k}$'s.

Two weight identities:

- $\sum_{k=m_0+1}^{m_0+m} w_{k,m} = 0$ (kills any constant offset, e.g. $\log a_0$)
- $\sum_{k=m_0+1}^{m_0+m} k w_{k,m} = 1$ (picks out the slope coefficient)

Useful consequence:

$$\sum_{k=m_0+1}^{m_0+m} w_{k,m}^2 = \frac{12}{m(m^2 - 1)} \asymp \frac{1}{m^3}.$$

The three-part split

Apply $\sum_k w_{k,m}(\cdot)$ to the per-scale identity:

$$\hat{\beta} = \sum_k w_{k,m} \left[\underbrace{\log a_0 + k\beta}_{\text{signal}} + \underbrace{a_1 \rho^{-k} + \phi(\rho^k) \rho^{-2k}}_{\text{det. bias}} + \underbrace{\log \bar{\xi}_k}_{\text{random}} \right].$$

Using the weight identity $\sum w = 0$ we kill $\log a_0$ and using $\sum kw = 1$ we pick out β . Hence $\hat{\beta} - \beta$ retains only the bias and random pieces.

Collect:

$$\hat{\beta} - \beta = \underbrace{\sum_k w_{k,m} (a_1 \rho^{-k} + \phi(\rho^k) \rho^{-2k})}_{R(m, m_0)} + \underbrace{\sum_k w_{k,m} \log(\bar{\xi}_k)}_{\text{random part}}.$$

Main theorem

Consider $\hat{\gamma}$ the log-log plot estimator for the critical exponent γ and an observable Y_i satisfying the following assumptions.

Assumptions (with $\xi_i := Y_i/\mathbb{E}Y_i$):

(A0) $Y_i > 0$.

(A1) $\mathbb{E}Y_i = a_0 i^\gamma \exp(a_1 i^{-1} + \phi(i) i^{-2})$ with ϕ bounded.

(A2) $\mathbb{E}|\xi_i|^{2+\delta}$ and $\mathbb{E}|\log \xi_i|^{2+\delta}$ are uniformly bounded in i (some $\delta > 0$).

(A3) $\sigma_k^2 := \text{Var}(\log \xi_k)$ satisfies $\sigma_k \rightarrow \sigma_\infty \in (0, \infty)$.

Theorem (Barbosa–Teixeira–Oliveira, 2026⁺)

Under (A0)–(A3), if $m_0 \rightarrow \infty$ and $nm \rightarrow \infty$ with $m/n \rightarrow 0$ (no separate growth requirement on m ; constant m is allowed), then

$$\sqrt{nm^3} (\hat{\gamma} - \gamma) \xrightarrow{d} \mathcal{N}\left(0, \frac{12 \sigma_\infty^2}{\log^2 \rho}\right).$$

Proof strategy

Recall the three-part split:

$$\hat{\beta} - \beta = R(m, m_0) + \sum_k w_{k,m} \log(\xi_k)$$

Taylor-expand the random piece. $\log \bar{\xi}_k = (\bar{\xi}_k - 1) + q(\bar{\xi}_k - 1)$.

$$\hat{\beta} - \beta = \underbrace{R(m, m_0)}_{\text{deterministic bias}} + \underbrace{S_{n,m}}_{\text{linear stochastic}} + \underbrace{Q_{n,m}}_{\text{quadratic remainder}}$$

where $S_{n,m} = \sum_k w_{k,m}(\bar{\xi}_k - 1)$ and $Q_{n,m} = \sum_k w_{k,m}q(\bar{\xi}_k - 1)$

Multiply by the standard deviation scale $\sqrt{nm^3}$ and treat each piece:

- $\sqrt{nm^3} S_{n,m} \xrightarrow{d} \mathcal{N}(0, 12\sigma_\infty^2)$ (Lindeberg–Feller)
- $\sqrt{nm^3} Q_{n,m} \xrightarrow{p} 0$ (good event + moment bound)
- $\sqrt{nm^3} R(m, m_0) \rightarrow 0$ (deterministic, controlled by m_0)

$S_{n,m}$ via Lindeberg–Feller

$S_{n,m}$ is the OLS slope applied to the centred stochastic part (with $\bar{\xi}_k = \bar{Y}_{\rho^k} / \mathbb{E} Y_{\rho^k}$, $\mathbb{E} \bar{\xi}_k = 1$):

$$S_{n,m} = \sum_{k=m_0+1}^{m_0+m} w_{k,m} (\bar{\xi}_k - 1).$$

A triangular array of **weighted i.i.d. averages**: n replicas inside each $\bar{\xi}_k$, m weighted terms in k .

Variance. Using independence across k (the Y_i 's at different scales are computed on disjoint annuli):

$$\text{Var}(S_{n,m}) = \frac{1}{n} \sum_k w_{k,m}^2 \text{Var}(\xi_k) \asymp \frac{1}{nm^3} \sigma_\infty^2.$$

Lyapunov $(2 + \delta)$ -condition. Under bounded $(2 + \delta)$ -th moments of ξ_k , we can show

$$\frac{1}{S_{n,m}^{2+\delta}} \sum_{k=m_0+1}^{m_0+m} \mathbb{E} \left[|w_{k,m} (\bar{\xi}_k - 1)|^{2+\delta} \right] \xrightarrow{m,n \rightarrow \infty} 0.$$

Lindeberg–Feller gives the CLT.

$Q_{n,m}$ Taylor remainder vanishes faster

On the **good event** $\mathcal{G}_k = \{\bar{\xi}_k \geq 1/2\}$, $q(\bar{\xi}_k - 1) = \log(\bar{\xi}_k) - (\bar{\xi}_k - 1)$ satisfies $|q(\bar{\xi}_k - 1)| \leq (\bar{\xi}_k - 1)^2$. So

$$|Q_{n,m}| \leq \sum_k |w_{k,m}| (\bar{\xi}_k - 1)^2 + \text{bad-event term.}$$

Good event. $\mathbb{E}(\bar{\xi}_k - 1)^2 \asymp \sigma_k^2/n$, so $\mathbb{E}|Q_{n,m}^{\text{good}}| \lesssim (1/n) \sum_k |w_{k,m}| \asymp 1/(nm)$. Multiplying by $\sqrt{nm^3}$ gives $\sqrt{m/n} \rightarrow 0$.

Bad event. Use union bound and Chebyshev to bound $\mathbb{P}(\mathcal{G}_k^c) \leq 4\sigma_k^2/n$. Then use Cauchy–Schwarz to bound $\mathbb{E}[Q_{n,m} \mathbb{1}_{\mathcal{G}_k^c}]$.

$R(m, m_0)$ finite-size bias — the geometric scale wins

The $R(m, m_0)$ term is **deterministic** bias from the linear regression:

$$R(m, m_0) = \sum_{k=m_0+1}^{m_0+m} w_{k,m} \left(a_1 \rho^{-k} + \phi(\rho^k) \rho^{-2k} \right).$$

Under reasonable assumptions both correction terms decay **geometrically** in k , and $\max_k |w_{k,m}| \asymp 1/m^2$. The leading ρ^{-k} term, summed against the symmetric weights, gives

$$R(m, m_0) \asymp \frac{\rho^{-m_0}}{m^2}.$$

Using just two points

Cost model. A single trial at scale $L = \rho^k$ costs $\asymp L^d$ operations, where d is the per-trial cost exponent of the simulator (e.g. $d = 2$ for percolation on the plane). With n replicas, the *simulation budget* is

$$B = n \rho^{(m_0+m)d}.$$

Take the two-point slope ($m = 2$). The smallest legal regression window — just two scales, the slope is the high-school formula

$$\hat{\gamma} = \frac{\log \bar{Y}_{\rho^{m_0+2}} - \log \bar{Y}_{\rho^{m_0+1}}}{\log \rho},$$

no OLS machinery needed. Balance bias against variance, saturate the budget:

$$m_0(B) = \left\lfloor \frac{1}{d+2} \log_{\rho} B \right\rfloor, \quad n(B) = \lfloor B / \rho^{(m_0+2)d} \rfloor.$$

Plug into the previous bias and variance bounds at $m = 2$ ($R \asymp \rho^{-m_0}$ and $\text{Var}(\hat{\gamma}) \asymp 1/n$):

$$\mathbb{E}[(\hat{\gamma} - \gamma)^2] \lesssim \underbrace{C_R \rho^{-2m_0}}_{\text{bias}^2} + \underbrace{C_S/n}_{\text{variance}} \lesssim C B^{-2/(d+2)}.$$

Question. Is the exponent $2/(d+2)$ *sharp*, or could a cleverer estimator do polynomially better?

Answer: $B^{-2/(d+2)}$ is sharp (up to logs)

Theorem (minimax lower bound)

Take $\mathcal{P} =$ distributions $\mathbb{W} = (\mathbb{W}_L)_{L \in \mathbb{N}}$ with $|\mathbb{E}\mathbb{W}_L - a_0 L^\gamma| \leq \exp(a_1/L)$ and $\|\mathbb{W}_L/\mathbb{E}\mathbb{W}_L\|_{2+\delta} < M$. Then for every $\log B \geq 2(d+2)/d$, there is a universal constant c such that

$$\inf_{\hat{\gamma}} \sup_{\mathbb{W} \in \mathcal{P}} \mathbb{E}_{\mathbb{W}}[(\hat{\gamma} - \gamma)^2] \geq \frac{(d+2)^2 c}{\log^2 B} B^{-2/(d+2)}.$$

Combined with the previous slide: the log-log plot is **minimax-rate-optimal up to a $\log^2 B$ factor** — no estimator can do polynomially better.

Closely related: [Carpentier & Kim'15] prove the analogous rate for the tail coefficient of *second-order Pareto* distributions, with essentially the same two-point KL construction.

Proof idea — two-point reduction

We adapt the **general two-point minimax lemma of Rogers' lecture notes [Rogers'18]** to our **log-log class** $\mathcal{P}$. Pick *two* candidate models $\mathbb{X}, \mathbb{Y} \in \mathcal{P}$ whose parameters γ differ by s . The minimax risk is lower-bounded by separating these two:

Two-point lemma

([Rogers'18], Thm. 1)

$$\inf_{\hat{\gamma}} \sup_{\mathbb{W} \in \mathcal{P}} \mathbb{E}_{\mathbb{W}}[(\hat{\gamma} - \gamma)^2] \geq \frac{1}{4} s^2 \exp(-\text{KL}(\mathbb{X} \parallel \mathbb{Y})).$$

Our construction (this is the new part). For each scale $L = \rho^k$,

$$X_L \sim e^Z, \quad Y_L \sim \begin{cases} e^Z & L < L^*, \\ e^{Z+\lambda_L} & L \geq L^*, \end{cases} \quad Z \sim \mathcal{N}(0, 1).$$

with $L^* = B^{1/(d+2)}$, $\lambda_L = \gamma \log L$, $\gamma = \log(1 - 1/L^*) / \log(1/L^*)$.

Direct check: $\mathbb{Y}$ stays inside $\mathcal{P}$ (the leading expansion holds at order L^{-1}), $\mathbb{E}X_L = 1$, and $\mathbb{E}Y_L = L^\gamma \cdot \mathbb{1}_{\{L \geq L^*\}} + \mathbb{1}_{\{L < L^*\}}$.

Proof idea — the “enemy” picks the best budget split

The KL between the two product measures factorises over scales:

$$\text{KL}(\mathbb{X} \parallel \mathbb{Y}) = \frac{1}{2} \sum_{L \geq L^*} n_L(B) \gamma^2 \log^2 L,$$

where $n_L(B)$ is the number of trials the statistician spends at scale L , subject to the budget $\sum_L n_L L^d \leq B$.

The enemy maximises the KL. Change of variable $M_L = n_L L^d / B$:

$$\max_{(M_L)} \sum_{L \geq L^*} B M_L L^{-d} \log^2 L \quad \text{s.t.} \quad \sum_L M_L \leq 1.$$

Linear in M_L on the simplex $\Rightarrow$ optimum at an extreme point: put all mass on the L that maximises $L^{-d} \log^2 L$. Calculus shows that this is L^* as soon as $\log B \geq 2(d+2)/d$.

Substituting $L^* = B^{1/(d+2)}$ and bounding $\log^2(1-x) \leq a^2 x^2$ gives $\text{KL} \leq a^2/2$, a *constant*. The $s^2 \cdot e^{-\text{KL}}$ in the two-point lemma then yields the announced rate.

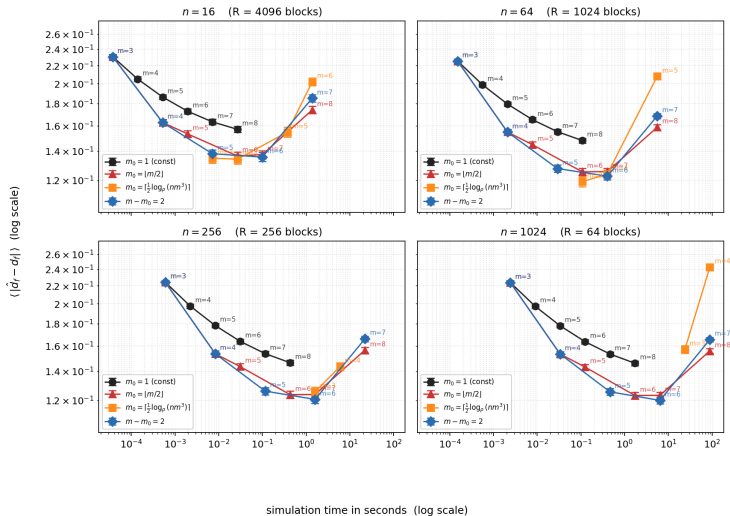
A note on what follows

Independence assumption is violated in the simulations

Theory (this talk and the article): the per-scale averages $\bar{Y}_{\rho^k}$ are built from **independent** samples — one fresh pool of n trials per scale k .

Simulations (next slide): we run **one** BFS pool and read off the cluster mass at *all* scales from each trial. The same realisation contributes to every $\bar{Y}_{\rho^k}$, so the per-scale averages are **correlated** across k within a trial.

MAE vs simulation time; one panel per n , four m_0 -schedules overlaid (square, $N = 8193$, pool = 65536)



= bfs pool elapsed = 1h34m28s trials = 65536 seed = 20260518 scales = [2, 4, 8, 16, 32, 64, 128, 256, 512, 1024, 2048, 4096] wall-time calibration: $c = 1.23e-08$ s/bu

Future work

- **Tune the best regime** in models with a *known* critical exponent (the fractal dimension d_f is our calibration target; do the same for the mean-cluster-size exponent γ , the percolation probability exponent β , ...).
- **Apply to models with unknown exponents.** Percolation in $d \geq 3$, random walks in random environment, polymers, interacting particle systems near their critical points — anywhere a log–log plot is currently read by eye.
- **Faster simulators.** Leath-style lazy BFS (for the cluster fractal dimension), GPU-parallel cluster generation.
- **Hypothesis test: “a critical exponent exists”.** In the spirit of Clauset–Shalizi–Newman power-law tests for heavy-tailed data, build a statistical procedure that rejects the power-law null when the log–log plot is incompatible with $\mathbb{E}Y_i \sim a_0 i^\gamma$.
- **Negative critical exponents.** $\gamma < 0$: $\mathbb{E}Y_i \rightarrow 0$, the $1/\log \bar{Y}$ and good-event/bad-event machinery change. Identify the corresponding moment assumptions and adapt the CLT.
- **Off-critical exponents.** Treat $\chi(p) \sim |p - p_c|^{-\gamma}$ and the β family: the regression variable is now $|p - p_c|$, not a length scale; budget allocates over a grid of p values rather than scales.

Thank you.

Questions?

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Code, slides and notes

<https://github.com/IgorSSBarbosa/impa-seminar-2026-05-18>